Final Exam

● Graded

Student

Yigeng Wang

Total Points

44.5 / 45 pts

Question 1

Overall Grades **44.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 0.5 pts** Point adjustment

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Yiqiang Wang	yw802	

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE:

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	4.5	5	5	5	5	5	5	5	44.5

Problem 1. [5] Find the general solution of the linear system:

$$\begin{aligned} 2x - 3y &= 1 \\ -2y - 3z &= 1 \\ x + 2z &= 1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & -2 & -3 & 1 \\ 2 & -3 & 0 & 1 \end{array}\right]$$

$R_3 - 2R_1 \to R_3$.
$R_2 \to -\frac{1}{2}R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & -\frac{5}{2} \end{array}\right] \xleftarrow{R_3 + 3R_2 \to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -3 & -4 & -1 \end{array}\right]$$

$R_2 + 3R_3 \to R_2$
$R_1 - 4R_3 \to R_1$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

$$\begin{cases} x = 11 \\ y = 7 \\ z = -5 \end{cases}$$

4.5/5

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(2)

(1) casting-out

$$\left[\begin{array}{ccc|c} 2 & 4 & 2 & 0 \\ -5 & 0 & 5 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \end{array}\right]$$

$R_2 \to \frac{1}{5}R_2$, $R_1 \to \frac{1}{2}R_1$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \end{array}\right]$$

$R_4 \to R_4 - R_1$, $R_2 \to R_1 + R_2$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

$R_1 \to R_1 - 2R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

rref is $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

pivot column is V_1 V_2.

basis of column space is

$\text{span}\left\{\begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix}\right\}$ because $\vec{V_3} = \begin{bmatrix} 2 \\ 5 \\ 1 \\ 2 \end{bmatrix} = \vec{V_2} - \vec{V_1}$

basis of row space is $\text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right\}$ +0.5

$x - z = 0$, $z = t \in R$.

$y + z = 0$

$y = t\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ $t \in R$.

basis of null space is

$\text{span}\left\{\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}\right\}$

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

A size is 2x3.

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = A\begin{bmatrix}0\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix} \qquad A\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}$$

$$T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = A\begin{bmatrix}1\\0\\-1\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix} \qquad A = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}^{-1}$$

$$T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = A\begin{bmatrix}0\\0\\-1\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix}$$

$$B = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix} \qquad \begin{bmatrix}0&1&-1\\1&0&1\\0&0&-1\end{bmatrix}$$

$\det(B) = -1 \cdot 1 = -1$

Cofactor of B = $\begin{bmatrix}0&-1&0\\-1&0&0\\0&0&-1\end{bmatrix}$ $\qquad B^{-1} = \begin{bmatrix}0&1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$

$(B)^T = \begin{bmatrix}0&1&0\\-1&0&0\\-1&1&-1\end{bmatrix}$

$B^{-1} =$

$$A = \begin{bmatrix}1&0&2\\1&2&2\end{bmatrix}\begin{bmatrix}0&-1&0\\1&0&0\\-1&-1&-1\end{bmatrix}$$

$$= \begin{bmatrix}2&0&2\\0&-2&-2\end{bmatrix} \qquad A = \begin{bmatrix}2&3&2\\4&3&2\end{bmatrix}$$

$$= \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix} \qquad A = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}$$

Problem 4. [5'] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

$\det(A-\lambda I)$

$$A-\lambda I = \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$$

$$(1-\lambda)\left[(1-\lambda)(2-\lambda)\right]$$

$$= (1-\lambda)^2(2-\lambda)$$

$\lambda_1 = \lambda_2 = 1$

$\lambda_3 = 2$

$$J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

for $\lambda_1 = \lambda_2 = 1$ AM = 2

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

eigenvector $\vec{V_1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

GM = 1

$(A-I)\vec{V_2} = \vec{V_1}$

$$\left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 1 \end{array}\right] \to \begin{array}{l} y = 0 \\ -2x = 1 \\ x = -\frac{1}{2} \end{array}$$

$$\vec{V_2} = \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$$

for $\lambda_3 = 2$.

$$\begin{bmatrix} -1 & -2 & 0 \\ 0 & -1 & 0 \\ -2 & -2 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 0 \\ 0 & -6 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$x - 2y = 0 \quad x = \frac{1}{3}z.$

$-6y - z = 0$

$6y = -z.$

$y = -\frac{1}{6}z.$

eigenvector $\vec{V_3} = \begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix}$

$$P = \begin{bmatrix} 0 & -\frac{1}{2} & \frac{1}{3} \\ 0 & 0 & -\frac{1}{6} \\ 1 & 0 & 0 \end{bmatrix} \quad P = \begin{bmatrix} 0 & -\frac{1}{2} & 2 \\ 0 & 0 & 1 \\ 1 & 1 & -6 \end{bmatrix}$$

$J = P^{-1}AP$

$P^{-1}AP$

$$A = \begin{bmatrix} 0 & 1 & 1 \\ -2 & 4 & 0 \\ 0 & -6 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{1}{3} \\ 0 & 0 & -\frac{1}{6} \\ 1 & 0 & 1 \end{bmatrix}$$

Problem 5. [5] Diagonalize the quadratic form

$$f(x, y, z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

we write the symmetric matrix of $f(x,y,z)$

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

$$f(x,y,z) = X^T A X$$

$$= (QY)^T A (QY)$$

$$= Y^T Q^T A Q Y$$

$$A - \lambda I = \begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = \det \begin{bmatrix} 3-\lambda & 3-\lambda & 0 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$$

applying row operation

$$= (3-\lambda)\left[(3-\lambda)(4-\lambda) - 1\right]$$

$$= (3-\lambda)\left[\begin{bmatrix} 1 & 1 & 0 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}\right]$$

$$= (3-\lambda)\left[(3-\lambda)(4-\lambda) - 1\right] - 1$$

$$= (3-\lambda)(\lambda^2 - 7\lambda + 10)$$

$$= (3-\lambda)(\lambda - 2)(\lambda - 5) = 0$$

$\lambda_1 = 3$
$\lambda_2 = 2$ different λ
$\lambda_3 = 5$. orthogonal eigenvector.

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{\sqrt{3}}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{\sqrt{3}}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix}$$

$$f(x,y,z) = f(u,v,w)$$

$$= 3u^2 + 4v^2 + 5w^2$$

for $\lambda_1 = 3$ eigenvector $\vec{V_1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ normalize $= \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

for $\lambda_2 = 2$ eigenvector $\vec{V_2} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ normalize $= \frac{1}{\sqrt{3}}\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

for $\lambda_3 = 5$ eigenvector $\vec{V_3} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ ~~normalize~~ $\frac{1}{\sqrt{6}}\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

$\vec{V_1} \cdot \vec{V_2}$ is perpendicular

$\vec{V_2}$ is perpendicular to $\vec{V_3}$ but not $\vec{V_1}$.

$$\vec{U_3} = \vec{V_3} - \frac{\vec{V_3} \cdot \vec{U_1}}{(u_1)^2} \vec{U_1} - \frac{\vec{V_3} \cdot \vec{U_2}}{(u_2)^2} \vec{U_2}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{3}{2}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{3}\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

normalize $\vec{U_3} = \frac{1}{\sqrt{6}}\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

$3v_1 - v_2 + v_3 = w$.

4/5

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & 8 \end{array}\right] \xrightarrow[R_1 \leftrightarrow R_3]{R_3 \to -\frac{1}{2}R_3} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 2 & -1 & 1 & 8 \\ 0 & 2 & -1 & -3 \end{array}\right]$$

$R_2 - 2R_1 \to R_2$

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & -1 & -3 \end{array}\right]$$

$R_3 + R_1 \to R_1$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & -1 & -3 \end{array}\right]$$

$R_3 \to R_3 - 2R_2$.

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & -3 \end{array}\right]$$

$c = 1$

$a + c = 4$

$b + c = 0$

$c = 1$

$a = 3$

$b = -1$

$c = 1$.

$\vec{w} = 3\vec{v_1} - \vec{v_2} + \vec{v_3}$.

Problem 7. [5'] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

apply Gram-schmidt

$\vec{u_1} = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$ normalize: $\vec{u_1} = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$\vec{u_2} = \vec{v_2} - \frac{\vec{v_2}\cdot\vec{u_1}}{\|u_1\|^2}\vec{u_1}$

$= \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{2}\vec{u_1}$

$= \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$ we normalize $\vec{u_2} = \frac{1}{\sqrt{3}}\begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

orthonormal basis for V

is

$$V = span\left\{\begin{bmatrix}0\\\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\end{bmatrix}, \begin{bmatrix}\frac{3}{\sqrt{15}}\\-\frac{1}{\sqrt{15}}\\\frac{1}{\sqrt{15}}\\-\frac{1}{\sqrt{15}}\end{bmatrix}\right\}.$$

Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

when $n=2$.

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\quad A = \begin{bmatrix} 0 & b \\ c & 0 \end{bmatrix}$ $\quad \det A = -cb = c^2 = b^2$.

it can be nonezero.

$A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} = \begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix}$

$a = -a = 0$
$d = -d = 0$.

$c = -b$
$b = -c$.

because of transpose diagonal doesn't change.
diag entries will be zero.

$A = \begin{bmatrix} 0 & \cdots & \cdots \\ \vdots & \ddots & \vdots \\ \cdots & \cdots & 0 \end{bmatrix}$ ($n-1$) $\quad tr(A) = 0 = \lambda_1 + \lambda_2 + \lambda_3 \cdots + \lambda_n = 0$.

$\det(A) = \lambda_1 \cdots \lambda_n$.

$A = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$ if $n=3$, 3 pairs

$\det(A) = abc - abc = 0$.

So I have to determine their is a $\lambda = 0$.

$\det(A^T) = \det(A) = \det(-A)$

$\det(A) = \det(A^T)$

$\det(-A) = (-1)^n \det(A)$

$\det(A) = (-1)^n \det(A)$

$[(-1)^n - 1] \det(A) = 0$

$n = \text{odd}$ $\quad \det(A) = 0$.

Problem 9. [5] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$$(\|\vec{u}+\vec{v}\|)^2 = (\|\vec{u}-\vec{v}\|)^2$$

$$\|\vec{u}\|^2+\|\vec{v}\|^2+2\vec{u}\cdot\vec{v} = \|\vec{u}\|^2+\|\vec{v}\|^2-2\vec{v}\cdot\vec{u}$$

$$4\,\vec{u}\cdot\vec{v} = 0$$

$$\vec{u}\cdot\vec{v} = 0$$

$\vec{u}$. $\vec{v}$ is perpendicular to each other.